HSBI — Bielefeld University of Applied Sciences and Arts

Quantitative Research Methods

Summer Semester 2026 · Examiner: Prof. Dr. Christoph Weisser

Final Mock Examination (Lectures 1–12)

Comprehensive coverage of ISLP Chapters 2–8, 10 and 13,
with emphasis on Chapters 7 (splines/GAMs), 8 (trees), 10 (deep learning) and 13 (multiple testing)

Duration:	**120 minutes**
Total credit:	**120 points**
Allowed aids:	non-programmable calculator; one handwritten A4 sheet of notes (both sides)

Name: ______________________ Matriculation no.: ______________

Problem	**P1**	**P2**	**P3**	**P4**	**P5**	**P6**	**P7**	**Total**
Maximum points	16	8	20	16	20	20	20	**120**
Achieved points								

Instructions

- Justify all answers. Numerical results without a derivation or a brief reasoning receive no credit.
- Round final numerical results to 3 decimal places unless stated otherwise. Small deviations caused by carrying rounded intermediate results are accepted.
- All numerical constants you need (powers of e, logarithms, $(0.95)^{20}$, t-quantiles) are provided in the problems.
- The number of points per problem roughly equals the recommended working time in minutes.

Problem 1: Moving beyond linearity — splines (16 points)

(a) **[4 P]** A *cubic spline* is a piecewise cubic polynomial that is continuous and has continuous first and second derivatives at each knot. How many degrees of freedom (free parameters, counting the intercept) does a cubic spline with K interior knots use? Justify your answer by counting the parameters of the polynomial pieces and the constraints at the knots, and evaluate the result for $K = 4$.

(b) **[4 P]** How many degrees of freedom does a *natural* cubic spline with the same K knots use (again evaluate for $K = 4$)? Which additional constraint does a natural spline impose, in which region of the predictor space does it act, and why is this constraint useful in practice?

(c) **[4 P]** A *smoothing spline* is the function g that minimises

$$\sum_{i=1}^{n} \big(y_i - g(x_i)\big)^2 + \lambda \int g''(t)^2 \, dt.$$

Explain the role of the two terms, and describe the behaviour of the solution $\hat{g}$ in the two limiting cases $\lambda \to 0$ and $\lambda \to \infty$.

(d) **[4 P]** Write down the *truncated-power basis* representation of a cubic spline with a single knot at ξ, define the truncated-power basis function explicitly, and explain in one sentence why adding this basis function does not destroy the smoothness of the fit at ξ.

Problem 2: Generalised additive models (8 points)

(a) **[3 P]** Write down the general form of a *generalised additive model* (GAM) for a quantitative response Y and predictors $X_1, \ldots, X_p$, and explain in one sentence how it extends multiple linear regression.

(b) **[3 P]** State one important *advantage* of GAMs that concerns the interpretation of the fitted functions f_j, and the main structural *limitation* of GAMs — including how this limitation can be repaired.

(c) **[2 P]** Describe in one or two sentences how the *backfitting* algorithm fits a GAM.

Problem 3: Regression and classification trees by hand (20 points)

An online platform records for $n = 6$ products the advertising budget x (in 1,000 EUR) and the weekly sales y (in 1,000 units):

Product i	1	2	3	4	5	6
Budget x_i	1	2	3	4	5	6
Sales y_i	2	4	3	3	8	10

A regression tree with a *single* split at cutpoint s partitions the data into $R_1 = \{i : x_i < s\}$ and $R_2 = \{i : x_i \geq s\}$ and predicts the region mean $\hat{y}_{R_m}$ in each region.

(a) **[8 P]** For each of the two candidate cutpoints $s = 2.5$ and $s = 4.5$, determine the two regions, the region means and the resulting total residual sum of squares $\mathrm{RSS}(s) = \sum_{m=1}^{2} \sum_{i \in R_m} (y_i - \hat{y}_{R_m})^2$. Which cutpoint does recursive binary splitting choose, and why?

(b) **[2 P]** Using the chosen split, predict the weekly sales for a new product with advertising budget $x = 5$.

(c) **[6 P]** A *classification* tree for a different task has a terminal node containing 8 training observations: 6 of class "Yes" and 2 of class "No". Compute the class proportions, the Gini index $G = \sum_k \hat{p}_k(1 - \hat{p}_k)$ and the entropy $D = -\sum_k \hat{p}_k \ln \hat{p}_k$ of this node *(use* $\ln 0.75 = -0.2877$ *and* $\ln 0.25 = -1.3863$*)*. Which class does the node predict?

(d) **[4 P]** In *cost-complexity pruning* one minimises, for a subtree $T \subseteq T_0$ of the large tree T_0,

$$\sum_{m=1}^{|T|} \sum_{i:\, x_i \in R_m} \left(y_i - \hat{y}_{R_m}\right)^2 + \alpha\, |T|.$$

Explain the role of the tuning parameter α: what happens for $\alpha = 0$, what happens as α increases, and how is α chosen in practice?

Problem 4: Ensembles — bagging, random forests, boosting (16 points)

(a) **[6 P]** Let $\hat{f}_1(x), \dots, \hat{f}_B(x)$ be B identically distributed (but not independent) tree predictions at a fixed point x, each with variance σ^2 and with pairwise correlation ρ. The variance of the bagged average is

$$\operatorname{Var}\Big(\tfrac{1}{B}\sum_{b=1}^{B}\hat{f}_b(x)\Big) = \rho\,\sigma^2 + \frac{1-\rho}{B}\,\sigma^2.$$

(i) Explain briefly why this formula shows that bagging reduces variance, and which term limits the reduction. (ii) Evaluate the variance for $B = 100$, $\rho = 0.6$ and $\sigma^2 = 4$, and compare it with the variance of a single tree. (iii) What value does the variance approach as $B \to \infty$?

(b) **[4 P]** How do *random forests* modify bagging at each split, and what is the recommended subset size m for classification as a function of p (evaluate for $p = 25$)? Explain, with reference to the formula in (a), why this modification improves on bagging.

(c) **[4 P]** Name the three tuning parameters of *boosting* for regression trees and state in one sentence each what they control. What is the "slow learning" idea behind boosting?

(d) **[2 P]** What does the *out-of-bag* (OOB) error of a bagged model estimate, and why is it available essentially for free?

Problem 5: Neural networks by hand (20 points)

Consider a tiny feed-forward network for a quantitative response with input $x = (x_1, x_2) = (1, 2)$, one hidden layer with two ReLU units ($g(z) = \max\{0, z\}$) and a linear output unit:

$$A_1 = g\big(-1 + 1\,x_1 + 2\,x_2\big), \qquad A_2 = g\big(-3 - 2\,x_1 + 1\,x_2\big), \qquad f(x) = 1 + 2\,A_1 + 4\,A_2.$$

(a) **[8 P]** Perform the forward pass step by step: compute both pre-activations, both activations A_1, A_2 and the network output $f(x)$. Comment briefly on what the ReLU did to the second unit and why such non-linear activations are essential.

(b) **[6 P]** A classification network with $K = 3$ classes produces the logits (output-layer pre-activations) $z = (z_1, z_2, z_3) = (1.0,\ 0.0,\ -1.0)$ for one observation whose true class is class 1. Compute all three softmax probabilities and the cross-entropy loss $-\ln \hat{p}_{\text{true}}$. *(Use $e^1 = 2.718$, $e^0 = 1.000$, $e^{-1} = 0.368$ and $\ln 4.086 = 1.408$.)*

(c) **[4 P]** Count the parameters of a fully connected network with $p = 4$ inputs, one hidden layer with $k = 3$ units and an output layer with $K = 2$ units, where every hidden and output unit has a bias. Show the count separately for the two weight layers.

(d) **[2 P]** A convolutional layer receives a 32×32 image and uses filters of size $F = 5$ with padding $P = 2$ and stride $S = 1$. Compute the spatial size of the output feature map from $\big(W - F + 2P\big)/S + 1$.

Problem 6: Multiple testing (20 points)

(a) **[4 P]** A lab performs $m = 20$ *independent* hypothesis tests, each at level $\alpha = 0.05$, and suppose all 20 null hypotheses are true. Define the family-wise error rate (FWER) and compute it for this situation *(use* $(0.95)^{20} = 0.3585$*)*. Interpret the result in one sentence.

(b) **[7 P]** A research group screens $m = 6$ candidate biomarkers; the ordered p-values are

j	1	2	3	4	5	6
$p_{(j)}$	0.001	0.004	0.011	0.030	0.045	0.230

Control the FWER at $\alpha = 0.05$ (i) with *Bonferroni* and (ii) with the *Holm step-down* procedure. Give the relevant thresholds, state for each method exactly which hypotheses are rejected, and explain why Holm can never reject fewer hypotheses than Bonferroni.

(c) **[6 P]** Apply the *Benjamini–Hochberg* (BH) procedure with FDR level $q = 0.05$ to the same six p-values: give the thresholds jq/m, determine which hypotheses are rejected, and state precisely what "FDR control at $q = 0.05$" guarantees for the resulting set of rejections.

(d) **[3 P]** A genomics group screens 20,000 genes in an *exploratory* study; the most promising hits will be re-tested in an independent follow-up experiment. Should they control the FWER or the FDR? Justify your recommendation.

Problem 7: Cumulative essentials (Chapters 2–6) (20 points)

(a) **[4 P]** True or false — justify briefly: "A more flexible statistical learning method always yields a lower *test* error than a less flexible one."

(b) **[4 P]** A logistic regression for loan default, $\log\left(\frac{p(x)}{1-p(x)}\right) = \beta_0 + \beta_1 x$ with x = number of missed payments, yields $\hat{\beta}_1 = 0.7$. Interpret the effect of *one additional missed payment* on the *odds* of default *(use* $e^{0.7} = 2.014$*)*. Why can the effect on the *probability* of default not be stated as a single number?

(c) **[4 P]** Give one reason to prefer k-fold cross-validation with $k = 5$ or 10 over *LOOCV*, and one reason to prefer it over a *single validation split*.

(d) **[2 P]** You have $p = 2{,}000$ predictors but expect only a handful to be truly related to the response. Ridge or lasso — which penalty is more appropriate, and why?

(e) **[6 P]** Match each scenario to exactly one method from {*linear regression*, *lasso*, *logistic regression*, *random forest*} (each method used exactly once) and justify each choice in one sentence:

(i) Estimate how much an additional square metre adds to the price of an apartment, with a confidence interval, from $n = 2{,}000$ sales.
(ii) Predict a continuous drug response from $p = 5{,}000$ gene-expression measurements for $n = 100$ patients; clinicians want a short list of relevant genes.
(iii) Predict whether a loan applicant will default (yes/no) and report to the regulator how each applicant characteristic changes the odds of default.
(iv) Predict machine failures from hundreds of sensor variables with strong non-linear effects and interactions; only predictive accuracy matters.

End of the examination — good luck!